Vakint reduction benchmark

Cached RustRed versus the FORM / FMFT reduction route

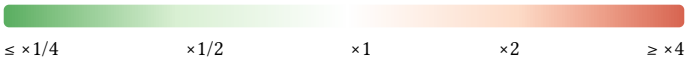
2026-10-24 · AMD EPYC 9754 128-Core Processor · CPU affinity 57

Reading guide. Bold values are RustRed times; FMFT times appear below. Multipliers are *RustRed ÷ FMFT*: green means faster, white parity, red slower. Both receive identical four-loop scalar inputs. The first physical denominator has power 2 (base) or 4 (held out); all others have power 1, auxiliaries 0.

Reduction time

Median seconds over 3 fresh processes per backend and case.

Topology	Base · power 2	Held out · power 4
H	**0.0359 s** FMFT 0.670 s ×0.054	**3.98 s** FMFT 7.37 s ×0.54
FG	**0.0450 s** FMFT 0.217 s ×0.21	**0.112 s** FMFT 0.430 s ×0.26
BMW	**0.0235 s** FMFT 0.181 s ×0.13	**0.997 s** FMFT 3.30 s ×0.30
X	**0.0822 s** FMFT 1.61 s ×0.051	**15.8 s** FMFT 21.0 s ×0.75



Method and correctness. Build profile: release (caller-declared); optimization/LTO settings are not inferred. rustc 1.98.1; FORM 5.0.0. Both retain masters in FMFT/MATAD normalization, with identical symbolic mass. Exact rational parity passed all eight cases at unit mass and renormalization scale, without truncating epsilon or using numerical tolerances.

Base cases hit prepared roots without applying rules. Held-out cases reuse cached descendants after root misses. Timing covers first public reduction calls, including FORM version checks, launch, I/O and parsing. Data loading, cache attachment, master evaluation and validation are excluded. Fresh processes run sequentially, alternating backends, with the recorded CPU affinity on a shared host. Filesystem pages are warm; this does not measure cold startup.

Selected validated run: `2026-10-24T06:59:51Z-8da99b9a5067`. Source: `8ef03d7fd755`. Data: `v0.1.2`. Samples and provenance: `data/results.json`. Refresh instructions: `../../AGENTS.md`.

Five-loop validation and benchmarks

640 declared, uncomputed masters · performance measurements pending

Historical DOT ingress acceptance

2026-10-23 · one sequential debug-build pass. This checks graph definitions; it does not time IBP reduction, master evaluation, package loading or cache preparation.

Validation workload	Recorded result
Master graphs / dotted / with numerator	640 / 441 / 204
Exact routing, full numerators, native ingress and DOT round trips	Passed
Complete graph-validation pass	61.8 s
Per graph: minimum / median / maximum	0.0443 / 0.0955 / 0.175 s
Numerical master values / prepared 5L cache in this run	Absent / absent

Physical powers, signed indices, mass terms, full numerators, loop-basis slots (including zero) and integration Jacobians were checked exactly. These checks establish the stored graph definitions, without claiming that the catalogue is an independent master basis. One observed run supports no statistical speedup claim. Earlier interrupted checks are not a timing baseline.

Five-loop reduction comparison

Retained-master workload	No persistent cache	With persistent cache
Complete quadratic UV fixture	Pending	Pending
Zero-momentum diagnostic	Pending	Pending
At least 12 held-out inputs	Pending	Pending

Required before filling these cells. Complete reductions and exact result/guard agreement; a completed, strict cache workload; three fresh-process runs per mode with fixed affinity and worker counts. Report graph preparation, loading, lowering, IBP, cache loading and end-to-end time separately. Cache size, preparation cost, intermediate reuse and held-out regressions accompany the comparison. No cache speedup is claimed yet.

Historical ingress: source `68f2263bf54d` (modified); RustRed `52c7463686b4` Symbolica `b6cca776f9dc`. Exact patch, executable, data and per-graph digests remain in JSON.

Latest validated integration cohort, separate from these historical timings: RustRed `91b72f8f5d42` Symbolica `9c8623ecb800`. Published upgrade: Symbolica `1205ec30890f`, FeynKit `d22e0f9fdbf8` final coordinated acceptance pending. Full revisions: `data/results.json`.